

DR ESLAM AHMED
eliteigcse.com

Algebraic Proof

Translate the words, expand and simplify, then factorise to show the property. Every proof is these three moves.

THE TOPIC ON ONE MAP

Algebraic Proof

01 The dictionary

$n, n + 1$ $2n$ $2n + 1$

02 Even, odd, multiple of k

$2(\dots)$ $8(\dots) + 2$

03 Square, cube, always positive

$(\dots)^2$ complete the square

04 Disprove

one counter-example

05 Proof by reasoning

consecutive integers primes

01 The Dictionary and the Three Moves

First turn the words into algebra. This table is your dictionary.

WHEN YOU SEE → YOU DO

“prove” / “show that ... for all integers n ”

→ use n — never test numbers

“consecutive”

→ $n, n + 1, n + 2$

“any two odd numbers”

→ two letters: $2a + 1, 2b + 1$

If the question says...

any integer

consecutive integers

an even number

an odd number

consecutive even / odd numbers

a multiple of k · one more than a multiple of k

a square number

you write...

n

$n, n + 1, n + 2$

$2n$

$2n + 1$

$2n, 2n + 2$ / $2n + 1, 2n + 3$

$kn \cdot kn + 1$

n^2

1

Translate

Use the dictionary. Two *different* odd numbers? Use two letters: $2a + 1, 2b + 1$.

2

Expand and simplify

Expand every bracket, collect like terms.

3

Factorise to prove

Make the answer *look like* what you must prove — then write one sentence of conclusion.

IDEA 01

Practise: The Dictionary and the Three Moves

My questions first, then selected practice.

Q1 **DR ESLAM** Use the dictionary

CH 2 · TOPIC 8 4 marks

Write an expression for (a) three consecutive integers (b) two consecutive even numbers (c) an odd number (d) a multiple of 3.

Q2 **DR ESLAM** Three moves

CH 2 · TOPIC 8 3 marks

Prove that the sum of any three consecutive integers is a multiple of 3.

SELECTED PRACTICE

Q3**PRACTICE**

Practice · Medium

CH 2 · TOPIC 8

4 marks

12 Prove that the mean of any four consecutive even integers is an integer.

(4 marks)**Q4****PRACTICE**

Practice · Medium

CH 2 · TOPIC 8

5 marks

13 (a) Prove that the sum of four consecutive whole numbers is always even.

(3 marks)

(b) Give an example to show that the sum of four consecutive integers is **not** always divisible by 4.

(2 marks)

Q5**PRACTICE**

Practice · Hard

CH 2 · TOPIC 8

4 marks

3 Prove that , for all positive values of n ,

$$\frac{(n+2)^2 - (n+1)^2}{2n^2 + 3n} = \frac{1}{n}$$

(4 marks)

Q6**PRACTICE**

Practice · Hard

CH 2 · TOPIC 8

4 marks

4 Prove algebraically that the difference between the squares of any two consecutive integers is equal to the sum of these two integers.

(4 marks)

Q7**PRACTICE**

Practice • Very Hard

CH 2 • TOPIC 8 5 marks

9 (a) Here are the first four terms of a sequence of fractions.

$$\frac{1}{1} \quad \frac{2}{3} \quad \frac{3}{5} \quad \frac{4}{7}$$

The numerators of the fractions form the sequence of whole numbers 1 2 3 4 ...

The denominators of the fractions form the sequence of odd numbers 1 3 5 7 ...

Write down an expression, in terms of n , for the n th term of this sequence of fractions.

(2 marks)

(b) Using algebra, prove that when the square of any odd number is divided by 4 the remainder is 1

(3 marks)

Q8**PRACTICE**

Practice • Very Hard

CH 2 • TOPIC 8 7 marks**12 (a)**

The diagram shows a cross placed on a number grid.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60

 L is the product of the left and right numbers of the cross. T is the product of the top and bottom numbers of the cross. M is the middle number of the cross.Show that when $M = 35$, $L - T = 99$.**(2 marks)****(b)** Prove that, for any position of the cross on the number grid above, $L - T = 99$.**(5 marks)**

02 Even, Odd, Multiple of k

The last line decides the mark: take out the right factor, then say why.

WHEN YOU SEE → YOU DO

“always even” → $2(\dots)$

“always odd” → $2(\dots) + 1$

“a multiple of 8” → $8(\dots)$

“remainder is 2 when divided by 8” → $8(\dots) + 2$

Even

$2 \times (\text{integer})$

take out 2

Odd

$2 \times (\text{integer}) + 1$

take out 2, leave +1

Multiple of k

$k \times (\text{integer})$

take out k

WORKED MODEL • MULTIPLE OF 8

Prove that $(2n + 3)^2 - (2n + 1)^2$ is a multiple of 8.

1 $(4n^2 + 12n + 9) - (4n^2 + 4n + 1)$

expand

2 $8n + 8$

simplify

3 $8(n + 1)$

factorise to prove

ANSWER $8(n + 1)$ — a multiple of 8, since $n + 1$ is an integer

× **TRAP** Showing it works for $n = 1, 2, 3$ is **not** a proof. Use n .

IDEA 02

Practise: Even, Odd, Multiple of k

My questions first, then selected practice, then real past papers.

Q9 DR ESLAM Multiple of 8

CH 2 · TOPIC 8 3 marks

Prove that $(n + 5)^2 - (n + 1)^2$ is a multiple of 8 for all integers n .

Q10 DR ESLAM Odd

CH 2 · TOPIC 8 3 marks

Prove that the product of two consecutive odd numbers is always odd.

Q11 DR ESLAM Multiple of 4

CH 2 · TOPIC 8 2 marks

Prove that the sum of two consecutive odd numbers is a multiple of 4.

SELECTED PRACTICE

Q12 **PRACTICE** Practice · Medium

CH 2 · TOPIC 8 3 marks

1 Prove algebraically that

$$(2n + 1)^2 - (2n + 1) \text{ is an even number}$$

for all positive integer values of n .

(3 marks)

Q13 **PRACTICE** Practice · Medium

CH 2 · TOPIC 8 4 marks

3 n is an integer greater than 1Prove algebraically that $n^2 - 2 - (n - 2)^2$ is always an even number.

(4 marks)

Q14 PRACTICE Practice · Hard

CH 2 · TOPIC 8 4 marks

2 Prove that the square of an odd number is always 1 more than a multiple of 4

(4 marks)

Q15 PRACTICE Practice · Hard

CH 2 · TOPIC 8 3 marks

12 c is a positive integer.

Prove that $\frac{6c^3 + 30c}{3c^2 + 15}$ is an even number.

(3 marks)

PAST PAPERS

Q16 PAST PAPER Edexcel 4MA1 · January 2020 · Paper 2HR · Q25

CH 2 · TOPIC 8 3 marks

25 N is a multiple of 5

$$A = N + 1$$

$$B = N - 1$$

Prove, using algebra, that $A^2 - B^2$ is always a multiple of 20

(Total for Question 25 is 3 marks)

Q17 PAST PAPER Edexcel 4MA1 · January 2020 · Paper 2H · Q17

CH 2 · TOPIC 8 3 marks

17 Prove that the difference between two consecutive square numbers is always an odd number.
Show clear algebraic working.

(Total for Question 17 is 3 marks)

Q18 **PAST PAPER** Edexcel 4MA1 · January 2021 · Paper 2HR · Q15**CH 2 · TOPIC 8** **4 marks**

15 Prove algebraically that the product of any two odd numbers is always an odd number.

(Total for Question 15 is 4 marks)

Q19 **PAST PAPER** Edexcel 4MA1 · May 2021 · Paper 1H · Q17**CH 2 · TOPIC 8** **3 marks**

17 Using algebra, prove that, given any 3 consecutive even numbers, the difference between the square of the largest number and the square of the smallest number is always 8 times the middle number.

(Total for Question 17 is 3 marks)

Q20**PAST PAPER**

Edexcel 4MA1 · January 2022 · Paper 1HR · Q15

CH 2 · TOPIC 8**3 marks**

- 15 Using algebra, prove that, given any 3 consecutive whole numbers, the sum of the square of the smallest number and the square of the largest number is always 2 more than twice the square of the middle number.

(Total for Question 15 is 3 marks)

Q21**PAST PAPER**

Edexcel 4MA1 · January 2022 · Paper 2H · Q18

CH 2 · TOPIC 8**3 marks**

- 18 Prove that when the sum of the squares of any two consecutive odd numbers is divided by 8, the remainder is always 2
Show clear algebraic working.

(Total for Question 18 is 3 marks)

Q22**PAST PAPER**

Edexcel 4MA1 · June 2025 · Paper 1HR · Q17

CH 2 · TOPIC 8**3 marks**

- 17 Prove that, for any three numbers which are consecutive multiples of 4, the difference between the square of the largest number and the square of the smallest number is always a multiple of 64

Show clear algebraic working.

(Total for Question 17 is 3 marks)

03 Square, Cube, Always Positive

To prove a square number, factorise into $(\quad)^2$. To prove “always positive”, complete the square.

WHEN YOU SEE → YOU DO

“is a square number”

→ factorise to $(\dots)^2$

“is a cube number”

→ simplify to $(\dots)^3$

“always positive”

→ complete the square

WORKED MODEL • A SQUARE NUMBER

Prove that $35 + (3x + 1)^2 - 2x(4x - 3)$ is a square number.

1 $35 + 9x^2 + 6x + 1 - 8x^2 + 6x$

expand

2 $x^2 + 12x + 36$

simplify

ANSWER $(x + 6)^2$ — a square number

WORKED MODEL • ALWAYS POSITIVE

Prove that $x^2 + x + 1$ is always positive.

1 $\left(x + \frac{1}{2}\right)^2 - \frac{1}{4} + 1$

complete the square

2 $\left(x + \frac{1}{2}\right)^2 + \frac{3}{4}$

simplify

ANSWER $\left(x + \frac{1}{2}\right)^2 \geq 0 \Rightarrow \text{expression} \geq \frac{3}{4} > 0$

IDEA 03

Practise: Square, Cube, Always Positive

My questions first, then selected practice.

Q23 DR ESLAM Factorise to a square

CH 2 · TOPIC 8 3 marks

Prove that $n(n + 2) + 1$ is always a square number.

Q24 DR ESLAM Complete the square

CH 2 · TOPIC 8 3 marks

Prove that $x^2 - 6x + 10 > 0$ for all values of x .

Complete the square first.

SELECTED PRACTICE

Q25

PRACTICE

Practice · Medium

CH 2 · TOPIC 8 4 marks

8 x is an integer.

Prove that $35 + (3x + 1)^2 - 2x(4x - 3)$ is a square number.

(4 marks)

Q26

PRACTICE

Practice · Hard

CH 2 · TOPIC 8 4 marks

8 n is the middle integer of three consecutive positive integers.

The three integers are multiplied to give a product.

n is then added to the product.

Prove that the result is a cube number.

(4 marks)

Q27**PRACTICE**

Practice · Hard

CH 2 · TOPIC 8 4 marks

- 9 Expressions for consecutive triangular numbers are

$$\frac{n(n+1)}{2} \text{ and } \frac{(n+1)(n+2)}{2}$$

Prove that the sum of two consecutive triangular numbers is always a square number.

(4 marks)**Q28****PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 8 3 marks

- 5 The product of two consecutive positive integers is added to the larger of the two integers.

Prove that the result is always a square number.

(3 marks)

Q29**PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 8 4 marks

10 The table gives information about the first six terms of a sequence of numbers.

Term number	1	2	3	4	5	6
Term of sequence	$\frac{1 \times 2}{2}$	$\frac{2 \times 3}{2}$	$\frac{3 \times 4}{2}$	$\frac{4 \times 5}{2}$	$\frac{5 \times 6}{2}$	$\frac{6 \times 7}{2}$

Prove algebraically that the sum of any two consecutive terms of this sequence is always a square number.

(4 marks)**Q30****PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 8 3 marks

11 Prove that $x^2 + x + 1$ is always positive.

(3 marks)

04 Disprove: One Counter-Example

To show a statement is **false**, one example is enough — but you must show the working that it fails.

WHEN YOU SEE → YOU DO

“show that ... is wrong” / “is Ali correct?”

→ one counter-example, with working

“always / sometimes / never true”

→ test positive, negative, zero

1

Try small values

Put $n = 1, 2, 3, \dots$ into the expression until one fails.

2

Show it fails

Not prime? Write its factors: $91 = 7 \times 13$. Not even? Show the number.

WORKED MODEL • “ALWAYS PRIME”?

$k = n^2 + 9n + 1$ is prime for $n = 1$ to 9?

1 $n = 6: 36 + 54 + 1 = 91$

try values

ANSWER $91 = 7 \times 13$ — not prime, so the statement is wrong

ALWAYS / SOMETIMES / NEVER Test a positive, a negative, and zero. One case each way = “sometimes”.

IDEA 04

Practise: Disprove: One Counter-Example

My questions first, then selected practice.

Q31 DR ESLAM Counter-example

CH 2 · TOPIC 8 2 marks

Sam says $n^2 + n + 11$ is prime for every positive integer n . Show that Sam is wrong.

Q32 DR ESLAM One example is enough

CH 2 · TOPIC 8 1 mark

Show that the statement “the sum of two prime numbers is always even” is false.

SELECTED PRACTICE

Q33**PRACTICE**

Practice · Medium

CH 2 · TOPIC 8 2 marks

6 $E = n^2 + n + 5$

Ali thinks that the value of E will be a prime number for any whole number value of n .

Is Ali correct?

You must give a reason for your answer.

(2 marks)

Q34**PRACTICE**

Practice • Medium

CH 2 • TOPIC 8 4 marks

7 p is a positive number. n is a negative number.

For each statement, tick the correct box.

	Always true	Sometimes true	Never true
$p + n$ is positive	☐	☐	☐
$p - n$ is positive	☐	☐	☐
$p^2 + n^2$ is positive	☐	☐	☐
$p^3 \div n^3$ is positive	☐	☐	☐

(4 marks)

Q35**PRACTICE**

Practice · Medium

CH 2 · TOPIC 8 1 mark

9 Which of these is a correct identity?

Circle your answer.

$x + 4x \equiv 5x$	$6x \equiv 18$	$2x + 1 \equiv 7$	$7x + 9 \equiv x$
--------------------	----------------	-------------------	-------------------

(1 mark)

Q36**PRACTICE**

Practice · Medium

CH 2 · TOPIC 8 3 marks

10

$$k = n^2 + 9n + 1$$

Mo says, “ k will be a prime number for all integer values of n from 1 to 9”

Show that Mo is wrong.

You **must** show that your value of k is **not** prime.

(3 marks)

Q37**PRACTICE**

Practice · Medium

CH 2 · TOPIC 8 1 mark

11 Tick whether the following statement is true or false.

Give a reason for your answer.

When n is a positive integer, the value of $2n$ is **always** a factor of the value of $20n$.

True ☐ False ☐

(1 mark)

05 Proof by Reasoning

Some proofs need a sentence, not just algebra: consecutive numbers, factors, primes.

WHEN YOU SEE → YOU DO

“prime”

→ a product of two factors > 1 is not prime

$n^3 - n, n(n+1)$

→ consecutive integers → even / multiple of 6

Two consecutive integers

one of them is even → the product is even

Three consecutive integers

one is even, one is a multiple of 3 → product is a multiple of 6

A product of two factors, both > 1

cannot be prime

$a^2 - b^2 = (a+b)(a-b)$ **is prime**

so $a - b = 1$: a and b are consecutive

WORKED MODEL • MULTIPLE OF 6

Prove that $n^3 - n$ is always a multiple of 6.

1 $n(n^2 - 1) = (n - 1)n(n + 1)$

factorise

ANSWER a multiple of 6

WHY $(n - 1), n, (n + 1)$ are three consecutive integers: one is even and one is a multiple of 3.

IDEA 05

Practise: Proof by Reasoning

My questions first, then selected practice.

Q38 **DR ESLAM** Square of an even number

CH 2 · TOPIC 8 2 marks

Prove that the square of any even number is a multiple of 4.

Q39 **DR ESLAM** Consecutive integers

CH 2 · TOPIC 8 2 marks

Prove that $n^2 + n$ is always even.

Factorise, then think about consecutive numbers.

SELECTED PRACTICE

Q40

PRACTICE

Practice · Hard

CH 2 · TOPIC 8

4 marks

6 (a) Show that $x(x - 1)(x + 1) = x^3 - x$

(1 mark)

(b) Prove that the difference between a whole number and the cube of this number is always a multiple of 6

(3 marks)

Q41

PRACTICE

Practice · Hard

CH 2 · TOPIC 8

2 marks

11 $a^2 - b^2 \equiv (a + b)(a - b)$

a and b are positive whole numbers with $a > b$

$a^2 - b^2$ is a **prime** number.

Why are a and b consecutive numbers?

(2 marks)

Q42**PRACTICE**

Practice · Hard

CH 2 · TOPIC 8**3 marks**

- 13** The lengths of the sides of a right-angled triangle are all integers. Prove that if the lengths of the two shortest sides are even, then the length of the third side must also be even.

(3 marks)**Q43****PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 8**3 marks**

- 1** (i) Factorise $2t^2 + 5t + 2$.

[2]

- (ii) t is a positive whole number.

The expression $2t^2 + 5t + 2$ can never have a value that is a prime number.
Explain why.

[1]**(3 marks)**

Q44 **PRACTICE** Practice · Very Hard

CH 2 · TOPIC 8 6 marks

3 Here are the first five terms of an arithmetic sequence.

7 13 19 25 31

Prove that the difference between the squares of any two terms of the sequence is always a multiple of 24.

(6 marks)

Q45 **PRACTICE** Practice · Very Hard

CH 2 · TOPIC 8 2 marks

4 Given that n can be any integer such that $n > 1$, prove that $n^2 - n$ is never an odd number.

(2 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|------------------------------------|--|
| 01 | Three moves | Translate (dictionary) → expand & simplify → factorise to prove. |
| 02 | Even / odd / k | $2() \cdot 2() + 1 \cdot k()$ — then one sentence. |
| 03 | Square • positive | Factorise to $()^2$. Always positive → complete the square. |
| 04 | Disprove | One counter-example, with the working that shows it fails. |
| 05 | Reasoning | Consecutive numbers, factors, primes — explain in words. |

BOTTOM LINE Translate → expand → simplify → factorise to prove → write the conclusion.

Answers

Algebraic Proof • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • THE DICTIONARY AND THE THREE MOVES

- Q1** Dr Eslam • Use the dictionary (a) $n, n + 1, n + 2$ (b) $2n, 2n + 2$ (c) $2n + 1$ (d) $3n$
- Q2** Dr Eslam • Three moves $n + (n + 1) + (n + 2) = 3n + 3 = 3(n + 1)$
- Q3** Practice • Medium $\text{mean} = \frac{8n + 12}{4} = 2n + 3$, an integer
- Q4** Practice • Medium (a) $4n + 6 = 2(2n + 3)$ (b) $1 + 2 + 3 + 4 = 10$
- Q5** Practice • Hard $\frac{2n + 3}{n(2n + 3)} = \frac{1}{n}$
- Q6** Practice • Hard $(n + 1)^2 - n^2 = 2n + 1 = n + (n + 1)$
- Q7** Practice • Very Hard (a) $\frac{n}{2n - 1}$ (b) $4(n^2 + n) + 1$
- Q8** Practice • Very Hard (a) $1224 - 1125 = 99$ (b) $(n^2 - 1) - (n^2 - 100) = 99$

IDEA 02 • EVEN, ODD, MULTIPLE OF K

Q9	Dr Eslam • Multiple of 8	$8n + 24 = 8(n + 3)$
Q10	Dr Eslam • Odd	$(2n + 1)(2n + 3) = 4n^2 + 8n + 3 = 2(2n^2 + 4n + 1) + 1$
Q11	Dr Eslam • Multiple of 4	$(2n + 1) + (2n + 3) = 4n + 4 = 4(n + 1)$
Q12	Practice • Medium	$4n^2 + 2n = 2(2n^2 + n), \text{ even}$
Q13	Practice • Medium	$4n - 6 = 2(2n - 3), \text{ even}$
Q14	Practice • Hard	$4n^2 + 4n + 1 = 4(n^2 + n) + 1$
Q15	Practice • Hard	$\frac{6c(c^2 + 5)}{3(c^2 + 5)} = 2c$
Q16	Edexcel 4MA1 • January 2020 • Paper 2HR • Q25	$4N, \text{ with } N = 5k: 20k$
Q17	Edexcel 4MA1 • January 2020 • Paper 2H • Q17	$2n + 1, \text{ odd}$
Q18	Edexcel 4MA1 • January 2021 • Paper 2HR • Q15	$2(2ab + a + b) + 1$
Q19	Edexcel 4MA1 • May 2021 • Paper 1H • Q17	$16n + 16 = 8(2n + 2)$
Q20	Edexcel 4MA1 • January 2022 • Paper 1HR • Q15	$2n^2 + 4n + 4 = 2(n + 1)^2 + 2$
Q21	Edexcel 4MA1 • January 2022 • Paper 2H • Q18	$8(n^2 + 2n + 1) + 2$
Q22	Edexcel 4MA1 • June 2025 • Paper 1HR • Q17	$(4n + 8)^2 - (4n)^2 = 64(n + 1)$

IDEA 03 • SQUARE, CUBE, ALWAYS POSITIVE

Q23 Dr Eslam • Factorise to a square

$$n^2 + 2n + 1 = (n + 1)^2$$

Q24 Dr Eslam • Complete the square

$$(x - 3)^2 + 1 \geq 1 > 0$$

Q25 Practice • Medium

$$x^2 + 12x + 36 = (x + 6)^2$$

Q26 Practice • Hard

$$(n - 1)n(n + 1) + n = n^3$$

Q27 Practice • Hard

$$(n + 1)^2$$

Q28 Practice • Very Hard

$$n(n + 1) + (n + 1) = (n + 1)^2$$

Q29 Practice • Very Hard

$$(n + 1)^2$$

Q30 Practice • Very Hard

$$\left(x + \frac{1}{2}\right)^2 + \frac{3}{4} > 0$$

IDEA 04 • DISPROVE: ONE COUNTER-EXAMPLE

Q31 Dr Eslam • Counter-example

$$n = 10: 121 = 11 \times 11, \text{ not prime}$$

Q32 Dr Eslam • One example is enough

$$2 + 3 = 5, \text{ which is odd}$$

Q33 Practice • Medium
 No: $n = 5$ gives $35 = 5 \times 7$

Q34 Practice • Medium
 sometimes • always • always • never

Q35 Practice • Medium

$$x + 4x \equiv 5x$$

Q36 Practice • Medium

$$n = 6: k = 91 = 7 \times 13$$

Q37 Practice • Medium
 True: $20n = 10 \times 2n$

IDEA 05 • PROOF BY REASONING

- Q38** Dr Eslam · Square of an even number
 $(2n)^2 = 4n^2 = 4(n^2)$
- Q39** Dr Eslam · Consecutive integers
 $n(n + 1)$: two consecutive integers, one of them even
- Q40** Practice · Hard
 (b) $(n - 1)n(n + 1)$: one even, one a multiple of 3
 prime $\Rightarrow a - b = 1$
- Q41** Practice · Hard
 $c^2 = 4(m^2 + n^2)$ even, so c even
- Q42** Practice · Hard
 (i) $(2t + 1)(t + 2)$ (ii) both factors ≥ 3
- Q43** Practice · Very Hard
 $12(a - b)(3a + 3b + 1)$; one factor even
- Q44** Practice · Very Hard
 $n(n - 1)$: consecutive, so even
- Q45** Practice · Very Hard